



Similarity preserving multi-task learning for radar target recognition

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ARTICLE INFO

Article history:

Received 7 July 2017

Revised 9 January 2018

Accepted 14 January 2018

Keywords:

Statistical recognition

Multi-task learning (MTL)

Transfer learning

Radar automatic target recognition (RATR)

High-resolution range profile (HRRP)

ABSTRACT

In this study, a statistical recognition model based on similarity preserving multi-task learning (SP-MTL) is developed for radar target recognition of high-resolution range profile (HRRP) data. A similarity preserving constraint, which describes the similarity information of HRRP samples, is introduced into multi-task learning to enhance the discriminative capability of the statistical model with limited training data. In addition, the SP-MTL model can be applied to the model prediction of new data based on transfer learning theory. Experiments on measured data show that the proposed model can achieve better recognition performance than traditional methods when training data is small. The application of the SP-MTL model to model prediction based on transfer learning theory can improve the learning precision of the new statistical model compared with the single-task learning of new data.

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1. Introduction

The work in radar automatic target recognition (RATR) area, according to the employment of the type of wideband radar signals, mainly focuses on high-resolution range profile (HRRP) recognition [5,6,10,14] and synthetic aperture radar (SAR) image or Inverse SAR (ISAR) image recognition [17,19,28,32,33]. A HRRP is composed of an amplitude of coherent summations of complex returns from target scatterers in each range cell, which contains abundant geometric structure information of the radar target, such as target size and scatterer distribution, on the radar target. Compared with the SAR and ISAR images, the HRRPs have been extensively studied and successfully applied to the RATR area with the property of easy acquisition and processing.

Several studies [6,10] have shown that the statistical recognition method is an important and efficient approach for RATR. Target-aspect sensitivity must be considered [9,12] when the HRRP is applied to statistical recognition. Therefore, a subset of multi-aspect HRRPs is collected from a target-aspect sector roughly without the motion through range cells (MTRC) of scatterers to address the abovementioned problem. Each subset from different target-aspect sectors is defined as an aspect-frame of the target, and a statistical model is developed for each aspect-frame [10]. The target membership of a measurement HRRP sample $\mathbf{x}$ is determined based on the posterior probabilities of all frames for all target memberships $\{p(c, m|\mathbf{x})\}_{c=1, m=1}^{C, M_c}$, where c denotes the target membership, C denotes the number of targets, m denotes the frame membership, and M_c denotes the number of frames for target c . In accordance with the Bayesian paradigm, $p(c, m|\mathbf{x}) \propto p(\mathbf{x}|c, m)p(c,$

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m), where $p(\mathbf{x}|c, m)$ is the frame-conditional likelihood of the m th frame for target c , and $p(c, m)$ is the prior frame probability. Prior frame probability is typically assumed to be uniformly distributed. Consequently, an estimation precision of the frame-conditional likelihood significantly affects the performance of statistical recognition.

Several parametric statistical models have been developed to describe the frame-conditional likelihood of the HRRP data, such as Gaussian model [15], Gamma model [6], the two distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [10], subspace approximation model [8], probability principal components analysis (PPCA) model [8,27], and factor analysis (FA) model [25]. Ref. [8] indicated that the FA model can precisely describe the statistical characteristics of the HRRP samples, thereby achieving favorable performance in statistical recognition.

However, complicated statistical models with many unknown parameters, such as the FA model, typically require a multitude of training data. In practice, collecting sufficient training data from each target-aspect in each target is difficult due to the sampling rate limitation. Subsequently, the learning precision of statistical models cannot be fully guaranteed, thus weakening recognition performance. Therefore, improving the recognition performance of statistical models with small-sized training data is critical. Multi-task learning (MTL), which is the sharing of structural elements or parameters among all related tasks, was applied in studies [18,31]. A MTL mechanism improves the estimation accuracy of parameters for each task by reducing the number of parameters to be estimated. Ref. [9] proposed a hidden Markov model (HMM)-based MTL algorithm for real HRRP data; the structure of this algorithm can avoid the requirement of large training data. However, the HMM algorithm, as a statistical model for sequence data, for a HRRP sample in [21] requires calculating state transition probabilities and emission likelihoods across range cells to derive the frame-conditional likelihood for this sample. Therefore, the computational speed during the test stage is relatively slower in the HMM than in other statistical models [6,8,10,15,25,27]. In addition, a FA-based MTL (MTL-FA) model for the HRRP data was developed in [11]. In MTL-FA model, a statistical model is developed for each frame and all frames sharing the same loading matrix and inferring different latent factors. Although MTL-FA model is superior to certain conventional statistical models, it still faces some challenges with limited training data. The MTL-FA model cannot achieve a satisfactory recognition performance when the number of training samples per frame is less than 70, as revealed in experiments on measured data in Section 4.4 of this study.

A similarity preserving (SP) constraint is introduced into the MTL-FA model to enhance the discriminative capability of learned statistical models with small-sized training data. Through the SP constraint, reconstruction samples within the same target remain similar, whereas the reconstruction samples for different targets are required to be more different. The Kullback–Leibler (KL) distance between two data distributions is inversely proportional to the average similarity of samples in the two datasets; therefore, the discriminative capability of the learned statistical models can be improved by constraining similarity information. Thus, the proposed MTL-FA with SP constraint, which is referred to as the SP-MTL model in this study, can not only reduce the dependence of learning precision on training data size but also improve the discriminative capability of the learned statistical models.

Another contribution of this paper is that the SP-MTL model can also be applied to the model prediction of new data based on transfer learning theory. For small amounts of data obtained from unknown aspect-frames of an existing target or a new target in real time, we prefer to utilize the knowledge from learned statistical models of original training data to assist in the model prediction of new data. In recent years, transfer learning has emerged as a new framework that is aimed at helping to improve the learning of the predictive function of a target task using knowledge from the source task [2,22]. On the basis of Transfer Learning Theory in this study, we transfer the knowledge of shared model parameters to construct new statistical models of new data observed in real time.

In this paper, we propose a HRRP target recognition method based on the SP-MTL model. Experiments on measured data show that the proposed SP-MTL model can achieve better recognition performance than traditional methods when size of the training data is small. In addition, compared with Single-Task Learning (STL) of new data, applying the SP-MTL model on transfer learning-based model prediction can improve the learning precision of new statistical models.

The rest of this paper is organized as follows. The FA model is reviewed in Section 2.1. Then, the proposed FA-based SP-MTL model is introduced in Section 2.2. The inference of the proposed model and classification scheme is derived in Sections 3.1 and 3.2, respectively, followed by presenting the model prediction based on transfer learning theory in Section 3.3. The experimental results of the measured HRRP data are presented in detail in Section 4. Lastly, the conclusion of this study is presented in Section 5.

2. Bayesian model construction

2.1. Review of the FA model

The FA model [25] is a probabilistic generative model that depicts high-dimensional observations distributed in low latent subspaces. A previous work [8] used the FA model to describe the statistical characterization of real HRRP samples. The subset of multi-aspect HRRPs was collected from a target-aspect sector roughly without the MTRC of scatterers due to the target-aspect sensitivity mentioned in Section 1. Each subset of different target-aspect sectors was defined as an aspect-frame of the target. Then, the FA model was developed for each aspect-frame with distinct parameters. The generative model for the i th D -dimensional HRRP sample in the m th frame for target c , where $i \in \{1, \dots, N_m\}$ with N_m denotes the number of the HRRP samples in this frame, $m \in \{1, \dots, M_c\}$ with M_c denotes the number of frames for target c , and $c \in \{1, \dots, C\}$ with C

denotes the number of targets, is represented by

$$\mathbf{x}_i^{(c,m)} = \mathbf{A}^{(c,m)} \mathbf{w}_i^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}^{(c,m)}, \quad (1)$$

where $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^D$ is the mean vector of the m th frame for target c , and $\mathbf{A}^{(c,m)} \in \mathbb{R}^{D \times K^*}$ is the loading matrix with K^* that denotes the number of latent factors, that is, typically $K^* \ll D$. The latent factor $\mathbf{w}_i^{(c,m)} \in \mathbb{R}^{K^*}$ is usually assumed Gaussian distributed as $\mathbf{w}_i^{(c,m)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{K^*})$. Noise vector is expressed as $\boldsymbol{\varepsilon}_i^{(c,m)} \in \mathbb{R}^D \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Lambda}^{-1})$, where $\boldsymbol{\Lambda}$ is a diagonal matrix. Thus, $\mathbf{x}_i^{(c,m)} \sim \mathcal{N}(\boldsymbol{\mu}^{(c,m)}, \boldsymbol{\Lambda} + \mathbf{A}^{(c,m)} \mathbf{A}^{(c,m)T})$. The number of latent factors K^* for each frame can be inferred via the Bayesian information criterion [8]. During the training phase, $\sum_{c=1}^C M_c$ FA models of all frames are learned independently based on the whole training dataset from all C targets. The above method is referred to as the FA-based STL model (STL-FA).

2.2. FA-based SP-MTL

2.2.1. MTL-FA model

In general, the FA model requires substantial training data to achieve robust estimations. However, acquiring sufficient samples for each frame is challenging in real applications. Thus, the learning precision of the statistical model may not be guaranteed. A previous study [11] proposed the MTL-FA model which shares the loading matrix across all frames from given targets, and models all related HRRP samples. Compared with STL-FA model in [8], the MTL mechanism reduces the number of parameters to be estimated, thus increasing the potential to accurately estimate the sub-loading matrix of each frame.

In addition, the number of latent factors K^* in the traditional FA model is preset and fixed. However, a fixed K^* may lead to overfitting or underfitting if the value is set to excessively high or low. In [20], beta process factor analysis (BPFA), which is a nonparametric FA model developed by John Paisley, used a beta process prior to automatically infer the number of latent factors. Another study [11] incorporated this nonparametric structure into the MTL-FA model. By introducing the beta process structure, the generative MTL-FA model of the i th ($i = 1, 2, \dots, N_m$) D -dimensional HRRP sample of the m th ($m = 1, 2, \dots, M_c$) frame for target c ($c = 1, 2, \dots, C$) can be represented as

$$\mathbf{x}_i^{(c,m)} = \mathbf{A} \mathbf{w}_i^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}_i^{(c,m)}, \mathbf{w}_i^{(c,m)} = \boldsymbol{\omega}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)}, \quad (2)$$

where $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_K] \in \mathbb{R}^{D \times K}$ is the shared loading matrix, $\mathbf{a}_k$ denotes the k th column of the loading matrix $\mathbf{A}$, K is preset to be relatively larger than the number of anticipated latent factors, $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^D$ is the mean vector of the m th frame that can be directly expressed as $\boldsymbol{\mu}^{(c,m)} = \frac{1}{N_m} \sum_{i=1}^{N_m} \mathbf{x}_i^{(c,m)}$, and $\boldsymbol{\varepsilon}_i^{(c,m)} \in \mathbb{R}^D$ is the noise vector.

For the MTL-FA model, the sub-loading matrix $\mathbf{A}^{(c,m)}$ is formed by selecting only a small subset from the columns in $\mathbf{A}$. The corresponding number of latent factors $K^{(c,m)}$ is also considerably smaller than K . To infer $K^{(c,m)}$ automatically, the latent factor $\mathbf{w}_i^{(c,m)} \in \mathbb{R}^K$ is decomposed into two parts: the weight vector $\boldsymbol{\omega}_i^{(c,m)}$ and the binary vector $\mathbf{z}_i^{(c,m)}$ in which each element is $\mathbf{z}_{i,k}^{(c,m)} \in \{0, 1\}$. Meanwhile, $\odot$ denotes an element-wise multiplication of two vectors. The BPFA can achieve desired sparseness by modeling $\mathbf{z}_{i,k}^{(c,m)}$ from a Bernoulli process parameterized by a beta process. Therefore, the appropriate number of latent factors $K^{(c,m)}$ for the m th frame for target c , which best fits the data, is determined automatically. Parameters in the hierarchical model are assumed to obey the following distributions:

$$\begin{aligned} \mathbf{a}_k &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D) \\ \boldsymbol{\omega}_i^{(c,m)} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\alpha}^{(c,m)-1} \mathbf{I}_K), \boldsymbol{\alpha}^{(c,m)} \sim \text{Gamma}(g_0, h_0) \\ \mathbf{z}_{i,k}^{(c,m)} &\sim \text{Bernoulli}(u_k^{(c,m)}), u_k^{(c,m)} \sim \text{Beta}(e_0, f_0) \\ \boldsymbol{\varepsilon}_i^{(c,m)} &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\delta}^{(c,m)-1} \mathbf{I}_D), \boldsymbol{\delta}^{(c,m)} \sim \text{Gamma}(a_0, b_0), \end{aligned} \quad (3)$$

where a_0, b_0, e_0, f_0, g_0 and h_0 are preset hyperparameters; $\text{Gamma}(\cdot)$, $\text{Bernoulli}(\cdot)$ and $\text{Beta}(\cdot)$ are referred to as Gamma, Bernoulli, and Beta densities, correspondingly, that are parameterized by their respective arguments within the parentheses.

2.2.2. SP constraint

The thought of MTL-FA model is to utilize shared information among different targets. Considering that MTL-FA model does not use the discriminative information [16]; and thus, we incorporate discriminative information into the MTL-FA model by introducing the SP constraint. We refer to the proposed model as the SP-MTL model.

The following expression is adopted to define the similarity between two HRRP samples:

$$s_{i,j}^{(c_1,m_1),(c_2,m_2)} = \frac{(\mathbf{x}_i^{(c_1,m_1)})^T (\mathbf{x}_j^{(c_2,m_2)})}{\|\mathbf{x}_i^{(c_1,m_1)}\|_2 \|\mathbf{x}_j^{(c_2,m_2)}\|_2}, \quad (4)$$

where $\mathbf{x}_i^{(c_1,m_1)}$ denotes the i th sample of the m_1 th frame for target c_1 , and $\mathbf{x}_j^{(c_2,m_2)}$ denotes the j th sample of the m_2 th frame for target c_2 . The more similar these two samples are, the higher the similarity between them. Moreover, the KL distance is

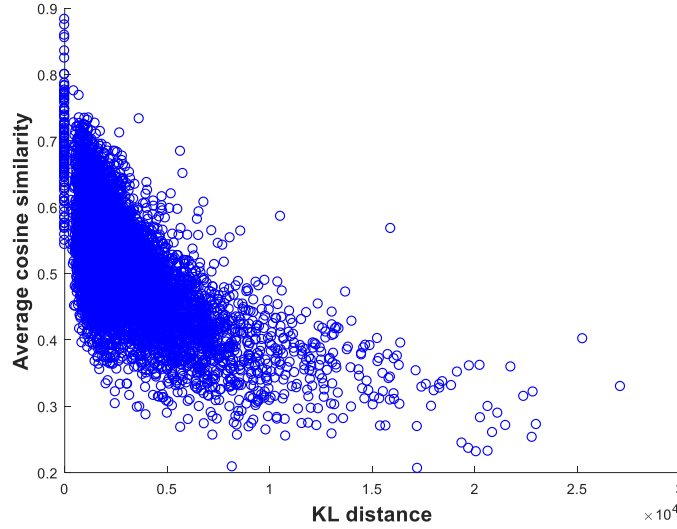


Fig. 1. Relationship between the KL distance of two frame-conditional PDFs and the average similarity of all samples from the two frames.

typically used to measure the closeness between statistical models [29] in pattern recognition. The KL distance of the two frame-conditional probability density functions (PDFs) can be represented as follows:

$$KL(p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1) || p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)) = \int (p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1) - p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)) \ln \frac{p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1)}{p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)} d\mathbf{x}, \quad (5)$$

where $p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1)$ and $p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)$ denote the frame-conditional PDFs of $\mathbf{x}_i^{(c_1, m_1)}$ and $\mathbf{x}_j^{(c_2, m_2)}$, and $KL(p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1) || p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2))$ measures the similarity between $p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1)$ and $p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)$. The proximity of the two PDFs to each other determines the KL distance between these PDFs. The closer of these two PDFs are, the smaller KL distance between them is. In addition, the KL distance satisfies $KL(p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1) || p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)) \geq 0$ with equality if and only if $p(\mathbf{x}_i^{(c_1, m_1)} | c_1, m_1) = p(\mathbf{x}_j^{(c_2, m_2)} | c_2, m_2)$. An experiment was conducted to search for a relationship between the KL distance of the PDFs and the average similarity of samples from two frames. In Fig. 1, the experimental result of the measured HRRP samples demonstrates that an approximately inverse proportion exists between the KL distance and average similarity. That is, average similarity decreases when the KL distance increases. This finding indicates that, if the average similarity of samples from two frames exhibit a generally decreasing trend, then the corresponding KL distance between the two frames will increase.

2.2.3. SP-MTL model

If $\mathbf{X} = \{\mathbf{x}_i^{(c, m)}\}_{c=1, m=1, i=1}^{C, M_c, N_m}$ denotes all training HRRP samples, then we define a similarity matrix $\mathbf{S}$ with each element represented as

$$S_{i,j} = \begin{cases} s_{i,j}^{(c_1, m_1), (c_2, m_2)}, & c_1 = c_2, \\ 0, & c_1 \neq c_2 \end{cases}, \quad (6)$$

where $S_{i,j}$ denotes the element of row i and column j of $\mathbf{S}$, and the definition of $s_{i,j}^{(c_1, m_1), (c_2, m_2)}$ is expressed in Eq. (4). If $c_1 = c_2$, then $\mathbf{x}_i^{(c_1, m_1)}$ and $\mathbf{x}_j^{(c_2, m_2)}$ belong to the same target. if $c_1 \neq c_2$, then $\mathbf{x}_i^{(c_1, m_1)}$ and $\mathbf{x}_j^{(c_2, m_2)}$ belong to different targets. Eq. (6) indicates that we retain the similarity of samples within the same target (i.e., unchanged values), whereas the similarity of samples in different targets is set to zero. For example, assume that $\mathbf{X} = [\mathbf{x}_1^{(1,1)}, \mathbf{x}_1^{(1,2)}, \mathbf{x}_1^{(2,1)}, \mathbf{x}_1^{(2,2)}, \mathbf{x}_1^{(3,1)}, \mathbf{x}_1^{(3,2)}]$, where only one sample is present in each frame; and $\mathbf{x}_1^{(1,1)}$ and $\mathbf{x}_1^{(1,2)}$ are the samples for Target 1; $\mathbf{x}_1^{(2,1)}$ and $\mathbf{x}_1^{(2,2)}$ are the samples for Target 2; and $\mathbf{x}_1^{(3,1)}$ and $\mathbf{x}_1^{(3,2)}$ are the samples for Target 3. Therefore, $\mathbf{S}$ can be defined as

$$\mathbf{S} = \begin{bmatrix} s_{1,1}^{(1,1), (1,1)} & s_{1,1}^{(1,1), (1,2)} & 0 & 0 & 0 & 0 \\ s_{1,1}^{(1,2), (1,1)} & s_{1,1}^{(1,2), (1,2)} & 0 & 0 & 0 & 0 \\ 0 & 0 & s_{1,1}^{(2,1), (2,1)} & s_{1,1}^{(2,1), (2,2)} & 0 & 0 \\ 0 & 0 & s_{1,1}^{(2,2), (2,1)} & s_{1,1}^{(2,2), (2,2)} & 0 & 0 \\ 0 & 0 & 0 & 0 & s_{1,1}^{(3,1), (3,1)} & s_{1,1}^{(3,1), (3,2)} \\ 0 & 0 & 0 & 0 & s_{1,1}^{(3,2), (3,1)} & s_{1,1}^{(3,2), (3,2)} \end{bmatrix}. \quad (7)$$

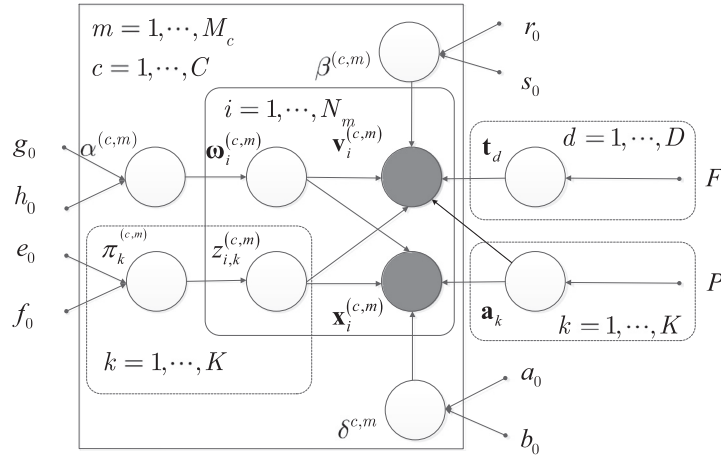


Fig. 2. Graphical representation of the SP-MTL model. The detailed generative process of which is described in Eqs. (2) and (11). In this figure, c is the target index, m is the frame index, i is the sample index in a frame, k is the latent factor index, and $\mathbf{x}_i^{(c,m)}$ is the i th HRRP sample in the m th frame for target c . The symbols C , M_c , N_m , K , and D at the edges indicate a stack of boxes for indexes c , m , i , k , and d , where C denotes the number of targets, M_c denotes the number of frames for target c , N_m denotes the number of samples in a frame, K denotes the number of latent factors, and D denotes the dimension of the HRRP samples.

Given a similarity matrix $\mathbf{S}$, the SP constraint term is defined as follows:

$$\min \|\mathbf{S} - (\mathbf{A}\mathbf{W} + \mathbf{M})^T (\mathbf{A}\mathbf{W} + \mathbf{M})\|_2^2, \quad (8)$$

where $\mathbf{W} = \{\mathbf{w}_i^{(c,m)}\}_{c=1, m=1, i=1}^{C, M_c, N_m}$ is a matrix, where each column is the latent factor $\mathbf{w}_i^{(c,m)}$, and $\mathbf{M} = \{\boldsymbol{\mu}^{(c,m)}\}_{c=1, m=1}^{C, M_c}$ is another matrix, where each column is the mean vector $\boldsymbol{\mu}^{(c,m)}$. The expression $\mathbf{A}\mathbf{W} + \mathbf{M}$ can be considered all HRRP construction samples. We expect the reconstruction samples within the same target to maintain a desirable similarity, whereas the similarity of the reconstruction samples in different targets are driven toward zero by minimizing the SP constraint term. The eigenvalue decomposition of similarity matrix $\mathbf{S}$ is considered

$$\mathbf{S} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{U}^T = \mathbf{U}\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\Sigma}^{\frac{1}{2}}\mathbf{U}^T, \quad (9)$$

where $\mathbf{U}$ denotes the eigenvector matrix of $\mathbf{S}$, and $\boldsymbol{\Sigma}$ denotes a diagonal matrix, where elements on the diagonal are eigenvalues that correspond to the eigenvectors in $\mathbf{U}$. If we define $\mathbf{V} = \boldsymbol{\Sigma}^{1/2}\mathbf{U}^T$, then Eq. (8) can be presented as

$$\min \|\mathbf{V} - \mathbf{T}(\mathbf{A}\mathbf{W} + \mathbf{M})\|_2^2, \quad (10)$$

where $\mathbf{T}$ is an orthogonal matrix with size $F \times D$, and $F = \sum_c M_c N_m$. Accordingly, we obtain the following SP constraint based on the MTL model for each sample:

$$\mathbf{v}_i^{(c,m)} = \mathbf{T}\mathbf{A}\mathbf{w}_i^{(c,m)} + \mathbf{T}\boldsymbol{\mu}^{(c,m)} + \boldsymbol{\eta}_i^{(c,m)}, \quad (11)$$

where $\mathbf{v}_i^{(c,m)}$, $\mathbf{T}\boldsymbol{\mu}^{(c,m)}$, and $\boldsymbol{\eta}_i^{(c,m)}$ correspond to $\mathbf{x}_i^{(c,m)}$, $\boldsymbol{\mu}^{(c,m)}$, and $\boldsymbol{\epsilon}_i^{(c,m)}$ in Eq. (2). Similar to noise vector $\boldsymbol{\epsilon}_i^{(c,m)}$, the error vector $\boldsymbol{\eta}_i^{(c,m)}$ is assumed to be the zero-mean Gaussian distribution $\boldsymbol{\eta}_i^{(c,m)} \sim \mathcal{N}(\mathbf{0}, 1/\beta^{(c,m)}\mathbf{I}_F)$. Parameter $\beta^{(c,m)}$ has a conjugate gamma prior $\beta^{(c,m)} \sim \text{Gamma}(r_0, s_0)$. The column of $\mathbf{T}$ follows a Gaussian prior similar to loading matrix $\mathbf{A}$,

$$\mathbf{t}_d \sim \mathcal{N}(\mathbf{0}, 1/F\mathbf{I}_F). \quad (12)$$

The Bayesian graph model of the SP-MTL model is illustrated in Fig. 2. Our proposed model incorporates discriminative information with a certain level of reconstruction capability by introducing the SP constraint into the MTL-FA model. The MTL-FA model cannot precisely describe the data when size of the training data are small. In such case, the SP constraint can help estimate the model parameters toward an appropriate direction. Thus, an improved recognition performance can be achieved by the SP-MTL model with small-sized training data.

3. Model learning and classification

3.1. Posterior distribution computation via variational Bayesian (VB) inference

The true distribution of latent variables can be usually inferred via Gibbs sampling using the Markov chain Monte Carlo method [13] and VB inference [4,7]. The VB inference, as a compromise between accuracy and efficiency, is adopted in this study.

In accordance with VB theory [3], a complete dataset is composed of observed data $\mathbf{D}$, latent variables, and parameters. The inference of the posterior for latent variables and parameters are inferred via the criterion for maximizing marginal likelihood (model evidence). For convenience, the set of all latent variables and parameters is denoted by $\mathbf{Z}$, which corresponds to the white circle in Fig. 2. Given the observed data $\mathbf{D}$ and hyperparameters θ , which are represented as a gray circle and a black point in Fig. 2, respectively, the marginal likelihood is defined as

$$\ln p(\mathbf{D}|\theta) = \int q(\mathbf{Z}) \ln p(\mathbf{D}|\mathbf{Z}, \theta) d\mathbf{Z}, \quad (13)$$

where $q(\mathbf{Z})$ is the unknown true distribution of $\mathbf{Z}$. The marginal likelihood in Eq. (13) can be further expressed as

$$\begin{aligned} \ln p(\mathbf{D}|\theta) &= \int q(\mathbf{Z}) \ln \frac{p(\mathbf{D}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \frac{q(\mathbf{Z})}{p(\mathbf{Z}|\mathbf{D}, \theta)} d\mathbf{Z} \\ &= \int q(\mathbf{Z}) \ln \frac{p(\mathbf{D}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} d\mathbf{Z} - \int q(\mathbf{Z}) \ln \frac{p(\mathbf{Z}|\mathbf{D}, \theta)}{q(\mathbf{Z})} d\mathbf{Z} \\ &= LB + KL(q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{D}, \theta)), \end{aligned} \quad (14)$$

where $LB = \int q(\mathbf{Z}) \ln \frac{p(\mathbf{D}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} d\mathbf{Z}$, which is the lower bound of marginal likelihood. $KL(q(\mathbf{Z})||p(\mathbf{Z}|\mathbf{D}, \theta))$ is the KL distance between a distribution $q(\mathbf{Z})$ and the exact posterior distribution of latent variables and parameters $p(\mathbf{Z}|\mathbf{D}, \theta)$, which is constantly positive and is equal to zero only when $q(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{D}, \theta)$. Therefore, variational inference can be considered a process of maximizing the lower bound. In such case, $q(\mathbf{Z}) = p(\mathbf{Z}|\mathbf{D}, \theta)$ and $LB = \ln p(\mathbf{D}|\theta)$. To obtain the optimal variational solution for $q(\mathbf{Z})$, we assume that latent variables and parameters are independent to obtain the optimal variational solution for $q(\mathbf{Z})$. Consequently, the joint distribution $q(\mathbf{Z})$ is approximated as a product of the marginal distribution of each latent variable and parameters according to mean field theory [23]:

$$q(\mathbf{Z}) = \prod_k q(\mathbf{Z}_k). \quad (15)$$

The variational method iteratively updates $q(\mathbf{Z}_k)$ alternately, hence the lower bound is maximized and approaches $\ln p(\mathbf{D}|\theta)$ until convergence.

In our model, observation data are $\mathbf{D} = \{\mathbf{x}_i^{(c,m)}, \mathbf{v}_i^{(c,m)}\}_{c=1, m=1, i=1}^{M_c, N_m}$, latent variables and parameters are $\mathbf{Z} = \{\alpha^{(c,m)}, u_k^{(c,m)}, \mathbf{a}_k, \delta^{(c,m)}, \mathbf{t}_d, \beta^{(c,m)}\}_{k=1, c=1, m=1, d=1}^{K, C, M_c, N_m, D}$, and hyperparameters are $\theta = \{a_0, b_0, e_0, f_0, g_0, h_0, r_0, s_0\}$. The derivation of updating formulas for the variational posterior that is implemented in the SP-MTL model is summarized as follows.

1) $q(\alpha^{(c,m)}) = \text{Gamma}(\alpha^{(c,m)}|g^{(c,m)}, h^{(c,m)})$ is assumed. The updating formulas for $q(\alpha^{(c,m)})$ are as follows:

$$\begin{aligned} g^{(c,m)} &= g_0 + \frac{1}{2} N_m K \\ h^{(c,m)} &= h_0 + \frac{1}{2} \sum_{i=1}^{N_m} \left\langle \left(\omega_i^{(c,m)} \right)^T \omega_i^{(c,m)} \right\rangle, \end{aligned} \quad (16)$$

where $\langle \cdot \rangle$ represents the expectation of the distribution of associated variables.

2) $q(\omega_i^{(c,m)})$ is expressed as a Gaussian distribution given the conjugate prior. The updating formulas of the covariance matrix and the mean vector are

$$\begin{aligned} q(\omega_i^{(c,m)}) &= \mathbb{N}(\omega_i^{(c,m)} | \tilde{\mathbf{w}}_i^{(c,m)}, \Sigma_i^{(c,m)}) \\ \Sigma_i^{(c,m)} &= \left(\langle \alpha^{(c,m)} \rangle \mathbf{I}_K + \langle \delta^{(c,m)} \rangle \left\langle \mathbf{z}_i^{(c,m)} (\mathbf{z}_i^{(c,m)})^T \right\rangle \odot \langle \mathbf{A}^T \mathbf{A} \rangle + \langle \beta^{(c,m)} \rangle \left\langle \mathbf{z}_i^{(c,m)} (\mathbf{z}_i^{(c,m)})^T \right\rangle \odot \langle (\mathbf{T} \mathbf{A})^T \mathbf{T} \mathbf{A} \rangle \right)^{-1} \\ \tilde{\mathbf{w}}_i^{(c,m)} &= \Sigma_i^{(c,m)} \text{diag} \left(\left\langle \mathbf{z}_i^{(c,m)} \right\rangle^T \right) \left(\langle \delta^{(c,m)} \rangle \langle \mathbf{A}^T \rangle \mathbf{x}_i^{(c,m)} + \langle \beta^{(c,m)} \rangle \langle \mathbf{A}^T \rangle \langle \mathbf{T}^T \rangle \mathbf{v}_i^{(c,m)} \right) \end{aligned} \quad (17)$$

3) $q(u_k^{(c,m)}) = \text{Beta}(u_k^{(c,m)} | e_k^{(c,m)}, f_k^{(c,m)})$ is assumed. Then,

$$\begin{aligned} e_k^{(c,m)} &= e_0 + \sum_{i=1}^{N_m} \langle z_{ik}^{(c,m)} \rangle \\ f_k^{(c,m)} &= f_0 + N_m - \sum_{i=1}^{N_m} \langle z_{ik}^{(c,m)} \rangle \end{aligned} \quad (18)$$

4) $q(z_{i,k}^{(c,m)})$ takes the form of a Bernoulli distribution. The updating formulas are as follows:

$$\begin{aligned} q(z_{i,k}^{(c,m)}) &= \text{Bernoulli} \left(z_{i,k}^{(c,m)} \middle| \frac{Ep_{i,k}^{(c,m)}}{Ep_{i,k}^{(c,m)} + Eb_{i,k}^{(c,m)}} \right) \\ Eb_{i,k}^{(c,m)} &\propto \exp \left(\ln (1 - u_k^{(c,m)}) \right) \end{aligned}$$

$$\begin{aligned}
Ep_{i,k}^{(c,m)} &\propto \exp \left\{ \left\langle \ln(u_k^{(c,m)}) \right\rangle \right. \\
&\quad - \frac{1}{2} \left[\langle w_{i,k}^{(c,m)2} \rangle \langle \delta^{(c,m)} \rangle \langle \mathbf{a}_k^T \mathbf{a}_k \rangle + \langle \beta^{(c,m)} \rangle \langle (\mathbf{T} \mathbf{a}_k)^T \mathbf{T} \mathbf{a}_k \rangle \right] \\
&\quad \left. + \left\langle (\boldsymbol{\omega}_i^{(c,m)})^T \right\rangle \langle (\mathbf{a}_k)^T \rangle \left(\psi_i^{(c,m)} + \phi_i^{(c,m)} \right) \right\},
\end{aligned} \tag{19}$$

where

$$\begin{aligned}
\psi_i^{(c,m)} &= \langle \delta^{(c,m)} \rangle \left\langle x_i^{(c,m)} - \sum_{r=1, r \neq k}^K \mathbf{a}_r z_{i,r}^{(c,m)} w_{i,r}^{(c,m)} \right\rangle \\
\phi_i^{(c,m)} &= \langle \beta^{(c,m)} \rangle \left(\langle \mathbf{T}^T \mathbf{v}_i^{(c,m)} \rangle - \langle \mathbf{T}^T \mathbf{T} \rangle \left\langle \sum_{r=1, r \neq k}^K \mathbf{a}_r z_{i,r}^{(c,m)} w_{i,r}^{(c,m)} \right\rangle \right).
\end{aligned} \tag{20}$$

5) The form of $q(\delta_k^{(c,m)})$ is similar to that of $q(\alpha^{(c,m)})$.

$$\begin{aligned}
q(\delta^{(c,m)}) &= \text{Gamma}(\delta^{(c,m)} | a^{(c,m)}, b^{(c,m)}) \\
a^{(c,m)} &= a_0 + \frac{1}{2} N_m D \\
b^{(c,m)} &= b_0 + \frac{1}{2} \sum_{i=1}^{N_m} \left\langle (\mathbf{x}_i^{(c,m)} - \mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)}))^T (\mathbf{x}_i^{(c,m)} - \mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)})) \right\rangle.
\end{aligned} \tag{21}$$

6) The form of $q(\beta_k^{(c,m)})$ is similar to that of $q(u_k^{(c,m)})$.

$$\begin{aligned}
q(\beta^{(c,m)}) &= \text{Gamma}(\beta^{(c,m)} | r^{(c,m)}, s^{(c,m)}) \\
r^{(c,m)} &= r_0 + \frac{1}{2} N_m F \\
s^{(c,m)} &= s_0 + \frac{1}{2} \sum_{i=1}^{N_m} \left\langle (\mathbf{v}_i^{(c,m)} - \mathbf{T} \mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)}))^T (\mathbf{v}_i^{(c,m)} - \mathbf{T} \mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)})) \right\rangle.
\end{aligned} \tag{22}$$

7) The updating formulas for the columns of loading matrix $\mathbf{a}_k$ are as follows:

$$\begin{aligned}
q(\mathbf{a}_k) &= \mathbb{N}(\mathbf{a}_k | \xi_k, \mathbf{B}_k) \\
\mathbf{B}_k &= \left[\langle \beta^{(c,m)} \rangle \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{i=1}^{N_m} \langle w_{i,k}^{(c,m)2} \rangle \langle z_{i,k}^{(c,m)} \rangle \langle \mathbf{T}^T \mathbf{T} \rangle + \langle \delta^{(c,m)} \rangle \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{i=1}^{N_m} \langle w_{i,k}^{(c,m)2} \rangle \langle z_{i,k}^{(c,m)} \rangle \mathbf{I}_D + D \mathbf{I}_D \right]^{-1}, \\
\xi_k &= \mathbf{B}_k \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{i=1}^{N_m} (\psi_i^{(c,m)} + \phi_i^{(c,m)})
\end{aligned} \tag{23}$$

where the definitions of $\psi_i^{(c,m)}$ and $\phi_i^{(c,m)}$ are expressed in Eq. (20).

8) To infer the posterior of $\mathbf{T}$, we furthermore have to write the expected complete likelihood as a sum of $\mathbf{T}$ rows to infer the posterior of $\mathbf{T}$. We denote $\mathbf{t}_f$ as a column vector that corresponds to the f th row in $\mathbf{T}$ and obtains

$$\begin{aligned}
q(\mathbf{T}) &= \prod_{f=1}^F q(\mathbf{t}_f) = \prod_{f=1}^F \mathbb{N}(\mathbf{t}_f | \boldsymbol{\varphi}_f, \mathbf{H}_f) \\
\mathbf{H}_f &= \left[\sum_{c=1}^C \sum_{m=1}^{M_c} \langle \beta^{(c,m)} \rangle \sum_{i=1}^{N_m} \left\langle (\mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)})) (\mathbf{A}(\mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)}))^T \right\rangle \right. \\
&\quad \left. + F \mathbf{I}_D \right]^{-1} \\
\boldsymbol{\varphi}_f &= \mathbf{H}_f \sum_{c=1}^C \sum_{m=1}^{M_c} \langle \beta^{(c,m)} \rangle \sum_{i=1}^{N_m} v_{i,f}^{(c,m)} \langle \mathbf{A} \rangle \langle \mathbf{w}_i^{(c,m)} \odot \mathbf{z}_i^{(c,m)} \rangle
\end{aligned} \tag{24}$$

In the succeeding experiments, we terminate the algorithm when the change of the lower bound LB between two iterations is smaller than 10^{-5} .

3.2. Classification

In this study, we predict the label of a test sample $\mathbf{x}^*$ via the maximum posterior criteria as follows:

$$\text{label}(\mathbf{x}^*) = \arg \max_c p(c, m | \mathbf{x}^*) \propto \arg \max_c p(c, m) p(\mathbf{x}^* | c, m). \tag{25}$$

The prior $p(c, m)$ is typically assumed to be uniformly distributed. The frame-conditional likelihood can be expressed as

$$p(\mathbf{x}^* | c, m) = \iiint \mathbb{N}(\mathbf{x}^* | \mathbf{A}(\mathbf{w}^{(c,m)}) + \boldsymbol{\mu}^{(c,m)}, (\delta^{(c,m)})^{-1} \mathbf{I}_D) q(\delta^{(c,m)}) q(\mathbf{w}^{(c,m)}) q(\mathbf{A}) d\delta^{(c,m)} d\mathbf{w}^{(c,m)} d\mathbf{A}, \tag{26}$$

where $\mathbf{w}^{(c,m)} = \{\mathbf{w}_i^{(c,m)}\}_{i=1}^{N_m}$ denotes the latent factors of the m th frame for target c . The posteriors $q(\delta^{(c,m)})$, $q(\mathbf{w}^{(c,m)})$, and $q(\mathbf{A})$ are estimated via variational inference in Section 3.1. However, calculating the integral of Eq. (26) is intractable. In general, we approximate Eq. (26) with the following expression:

$$p(\mathbf{x}^*|c, m) = \int \mathbb{N}(\mathbf{x}^*|\langle \mathbf{A} \rangle (\mathbf{w}^{(c,m)}) + \boldsymbol{\mu}^{(c,m)}, \langle \delta^{(c,m)} \rangle^{-1} \mathbf{I}_p) q(\mathbf{w}^{(c,m)}) d\mathbf{w}^{(c,m)}, \quad (27)$$

where $\langle \mathbf{A} \rangle$ and $\langle \delta^{(c,m)} \rangle$ are the posterior mean values of $\mathbf{A}$ and $\delta^{(c,m)}$, respectively. We assume that the posterior PDF $q(\mathbf{w}^{(c,m)})$ is a Gaussian distribution according to [12]:

$$\begin{aligned} q(\mathbf{w}^{(c,m)}) &= \mathbb{N}(\mathbf{w}_i^{(c,m)} | \boldsymbol{\gamma}^{(c,m)}, \mathbf{J}^{(c,m)}) \\ \boldsymbol{\gamma}^{(c,m)} &= \frac{1}{N_m} \sum_{i=1}^{N_m} \langle \mathbf{w}_i^{(c,m)} \rangle \\ \mathbf{J}^{(c,m)} &= \frac{1}{N_m} \sum_{i=1}^{N_m} \left((\mathbf{w}_i^{(c,m)} - \boldsymbol{\gamma}^{(c,m)}) (\mathbf{w}_i^{(c,m)} - \boldsymbol{\gamma}^{(c,m)})^T \right) \end{aligned} \quad (28)$$

Thus, the following frame-conditional likelihood is obtained:

$$\begin{aligned} p(\mathbf{x}^*|c, m) &= \mathbb{N}(\mathbf{x}^* | \mathbf{v}^{(c,m)}, \boldsymbol{\Gamma}^{(c,m)}) \\ \mathbf{v}^{(c,m)} &= \langle \mathbf{A} \rangle \boldsymbol{\gamma}^{(c,m)} + \boldsymbol{\mu}^{(c,m)} \\ \boldsymbol{\Gamma}^{(c,m)} &= \langle \mathbf{A} \rangle \mathbf{J}^{(c,m)} \langle \mathbf{A}^T \rangle + \langle \delta^{(c,m)} \rangle^{-1} \mathbf{I}_p \end{aligned} \quad (29)$$

Finally, we predict the label of test sample $\mathbf{x}^*$ via Eqs. (25) and (29).

3.3. Model prediction based on transfer learning

In many real-world applications, new data may be obtained in real time from certain unknown aspect-frames of existing or new targets. If information on existing training data is unavailable, then the corresponding statistical model cannot be learned during the training stage. To resolve this problem, new data can be integrated into training data to construct the accurate statistical model with the SP-MTL model. In such case, the statistical models must be rebuilt from zero for all data, which are computationally time-consuming in satisfying the requirements of model prediction in a minimal amount of time. Moreover, using only new data to develop model prediction is unreliable because the size of new data may be small. In recent years, transfer learning has emerged as a new framework for using knowledge from one or more source tasks (T_S) when developing a hypothesis for a new target task (T_T) [22,26]. Therefore, transfer learning theory is utilized in this study to transfer the shared knowledge of statistical models from T_S to the learning of T_T ; this transfer is aimed at assisting the model prediction of new data. Here, T_S denotes the statistical models learned by the original training data, whereas T_T denotes the model to be predicted based on new data. The HRRP data in T_S and T_T are denoted by D_S and D_T , correspondingly.

A critical problem in transfer learning is to determine what to transfer, which plays a key role in implementing transfer learning. A previous study [24] empirically proved that a negative transfer occurs when target data and tasks are too dissimilar with the source. The study claimed that such a phenomenon would result in transfer learning contributes to the reduced performance of learning in T_T . In case of model prediction, D_T is obtained from unknown aspect-frames that belong to the same class, and data in different frames within the same class are relevant. Moreover, measurement conditions for the HRRP data for each class are approximately the same in general. Thus, the conclusion that D_S and D_T are related to each other can be drawn. In addition, we choose to transfer model parameters $\mathbf{A}$ and $\mathbf{T}$ in T_S to T_T . For the proposed SP-MTL model, all frames shared a joint loading matrix $\mathbf{A}$, and the sub-loading matrix $\mathbf{A}^{(c,m)}$ for each frame is formed by selecting only a small subset of columns in $\mathbf{A}$, thereby indicating that frames are related by sharing low-dimensional representations. Thus, D_S and D_T are expected to exhibit similar properties if the shared model parameters learned in T_S are transferred. If D_S and D_T are highly dissimilar, then an accurate description of D_T cannot be guaranteed by transfer learning, which may lead to a negative transfer. From the preceding discussion, D_S is determined to be related to D_T , and the relevance of data can guarantee that T_S will become an improved learner by using transferred knowledge.

If data observed in real time are denoted by $\mathbf{X}^{(c^*,m^*)} = [\mathbf{x}_1^{(c^*,m^*)}, \mathbf{x}_2^{(c^*,m^*)}, \dots, \mathbf{x}_i^{(c^*,m^*)}, \dots, \mathbf{x}_{N_m^*}^{(c^*,m^*)}]$, then SP-MTL model prediction for each sample based on transfer learning can be presented as

$$\begin{cases} \mathbf{x}_i^{(c^*,m^*)} = \mathbf{A}^{(S)} \mathbf{w}_i^{(c^*,m^*)} + \boldsymbol{\mu}^{(c^*,m^*)} + \boldsymbol{\varepsilon}_i^{(c^*,m^*)} \\ \mathbf{v}_i^{(c^*,m^*)} = \mathbf{T}^{(S)} \mathbf{A}^{(S)} \mathbf{w}_i^{(c^*,m^*)} + \mathbf{T}^{(S)} \boldsymbol{\mu}^{(c^*,m^*)} + \boldsymbol{\eta}_i^{(c^*,m^*)} \end{cases} \quad (30)$$

The learning of T_T no longer updates shared model parameters $\mathbf{A}$ and $\mathbf{T}$. Alternatively, we transfer $\mathbf{A}^{(S)}$ and $\mathbf{T}^{(S)}$ learned from the original training data to T_T . This transfer indicates that the corresponding $\mathbf{A}^{(c,m)}$ and $\mathbf{T}^{(c,m)}$ of each aspect-frame for new data are formed by selecting a small subset of columns in $\mathbf{A}^{(S)}$ and $\mathbf{T}^{(S)}$. In addition, $\boldsymbol{\mu}^{(c^*,m^*)} = \frac{1}{N_m^*} \sum_{i=1}^{N_m^*} \mathbf{x}_i^{(c^*,m^*)}$. The new model parameters, $\mathbf{Z}^{c^*,m^*} = \{\alpha^{c^*,m^*}, u_k^{c^*,m^*}, \delta^{c^*,m^*}, \beta^{c^*,m^*}\}$, have the same priori distribution assumption as the corresponding parameters in T_S and are estimated according to updating formulas in Eqs. (16), (18), (21), and (22). Model prediction based on transfer learning transfers the shared model parameters from T_S to T_T which not only efficiently saves

Table 1
Parameters of the planes and radar in the recognition experiment.

Radar parameters	Center frequency		5520 MHz
	Bandwidth		400 MHz
Planes	Length (m)		Height (m)
	Width (m)		
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

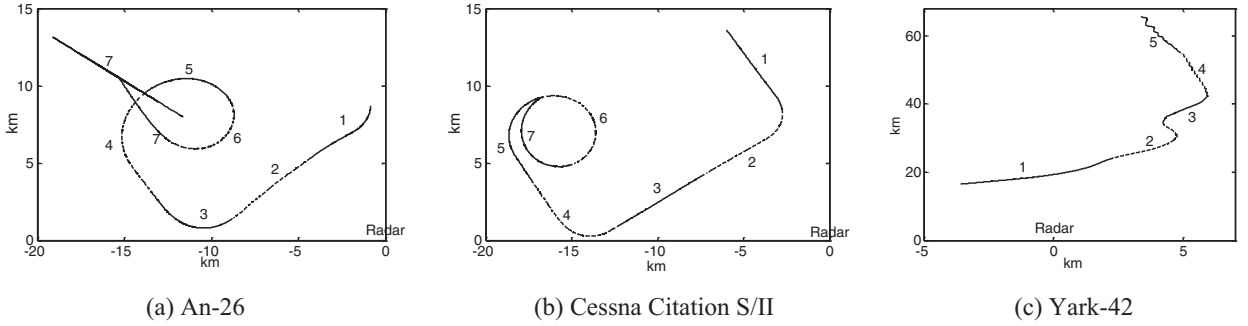


Fig. 3. Projections of target trajectories onto the ground plane.

learning time but also develops a more accurate hypothesis for a small amount of new data compared with model prediction without transferred knowledge.

4. Experimental results

4.1. Measured data

We examine the performance of the proposed model on measured HRRP data from three real airplanes used in [8, 30]. The parameters of the airplane targets and measurement radar are listed in Table 1, and the projections of target trajectories onto the ground plane are show in Fig. 3. To select the appropriate training data and test data, the measured data are segmented as follows: An-26 and Cessna Citation S/II are segmented into 7 segments; Yark-42 is segmented into 5 segments. With the Exception of the fifth segment of Yark-42, which contains 10,000 HRRP samples, each segment of three real airplanes contains 25,600 HRRP samples and each sample is a 256-length vector, that is, $D=256$. To test the generalization performance of the recognition methods, there are two requirements for choosing the training data and test data. (a) The Training data cover almost all of the target-aspect angles of the test data. (b) The Elevation angles of the test data are different from those of the training data. Therefore, data from the second segment and apart of the fifth segment of Yark-42, the sixth and seventh segments of Cessna Citation S/II, and the fifth and sixth segments of An-26 basically meet the requirements for training data, which are taken as the training samples and other data segments are taken as the test samples in our experiments. A total of 35,840 HRRP samples from Yark-42, 51,200 from Cessna Citation S/II, and 51,200 from An-26 are used as training data.

To overcome the problem of target-aspect sensitivity, the subset of multi-aspect HRRPs is collected from a target-aspect sector roughly without the MTRC. The limitation of target-aspect change for avoiding the MTRC of scatterers is expressed by

$$\delta\varphi \leq (\delta\varphi)_{\text{MTRC}} = \Delta R/L, \quad (31)$$

where ΔR and L are the resolution of the range cell and the maximum target dimension in cross range, respectively. Since the scatterers are localized immovably in each range cell within $(\delta\varphi)_{\text{MTRC}}$, the scattering center model almost remains unchanged. Each subset from a target-aspect sector is defined as an aspect-frame, and each frame has 1024 samples. The number of frames is 35, 50, and 50 for the targets Yark-42, Cessna Citation S/II, and An-26, correspondingly. Moreover, each sample is a 256-length vector. Thus, $C=3$, $M_1=35$, $M_2=M_3=50$, $D=256$, and $N_{M_1}=N_{M_2}=N_{M_3}=1024$. In the following experiments, we train the proposed model with different N_m values (i.e., different training samples per frame).

Similar to [8], we address the amplitude scale sensitivity by normalizing each HRRP sample by dividing its L_2 norm. Moreover, due to the fact that an HRRP sample is only a part of the received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRRP sample can vary with the measurement. This scenario is considered a time-shift sensitivity problem. To overcome this problem, all HRRP samples in one frame are slide correlation processed, and then, the HRRP samples in one frame are shifted with the same number of range cells to retain the geometric center of the average HRRP at the 128th range cell.

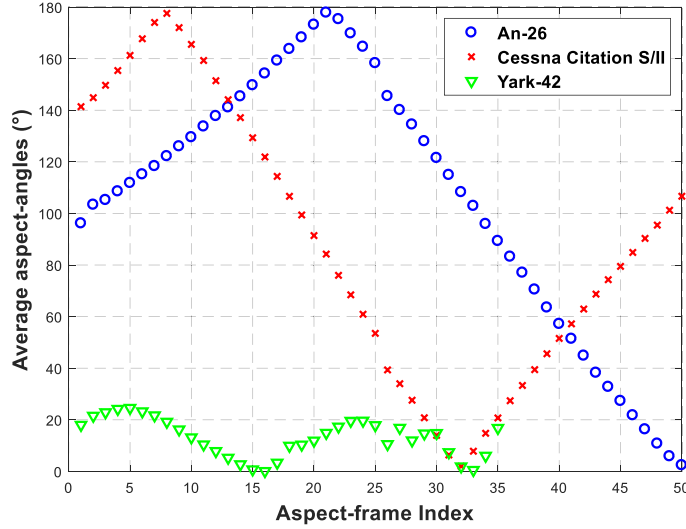


Fig. 4. Average aspect-angles of each frame for the three targets.

4.2. Similarity matrix setting analysis

We expect the discriminative information to assist in the learning of the generative model by introducing the SP constraint into the MTL-FA model. The key aspect of the SP constraint is its setting of the similarity matrix because different similarity matrix settings will have varying effects on model performance. Therefore, we make an analysis on different similarity matrices and compare their performance in this subsection.

First, we define that if the absolute value of the difference between the average aspect-angles of two aspect-frames for different targets is less than or equal to 1.5° , then the two aspect-frames can be considered approaching aspect-frames. Otherwise, the two aspect-frames are regarded as dissimilar aspect-frames. This condition can be presented by the following mathematical expression:

$$(m_i, m_j) \begin{cases} (m_i, m_j) \in m_{appro}, & \text{if } |w_i - w_j| \leq 1.5^\circ \\ (m_i, m_j) \in m_{dis}, & \text{if } |w_i - w_j| > 1.5^\circ \end{cases} \quad (32)$$

where w_i and w_j denote the average aspect-angles of the i th aspect-frame m_i for target c_1 and the j th aspect-frame m_j for target c_2 , respectively, as c_1 and c_2 are different targets; $(m_i, m_j) \in m_{appro}$ denotes that m_i and m_j are classified as the approaching aspect-frames for different targets; and $(m_i, m_j) \in m_{dis}$ denotes that m_i and m_j are classified as the dissimilar aspect-frames for different targets. The average aspect-angles of each frame are shown in Fig. 4, where the horizontal axis denotes the index of each aspect-frame for each target (i.e., An-26 and Cessna Citation S/II contain 50 aspect-frames, whereas Yark-42 contains 35 aspect-frames). In Fig. 4, the average aspect-angle of the 10th aspect-frame for An-26 is 129.54° , whereas that of the 15th aspect-frame for Cessna Citation S/II is 129.33° . The absolute difference of both aspect-frames is 0.21° . Thus, the two aspect-frames are classified as approaching aspect-frames for different targets.

Second, we define three similarity matrices by considering the information of aspect-angles.

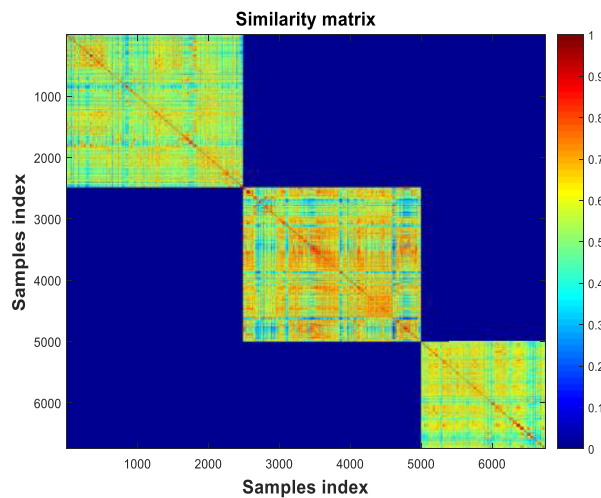
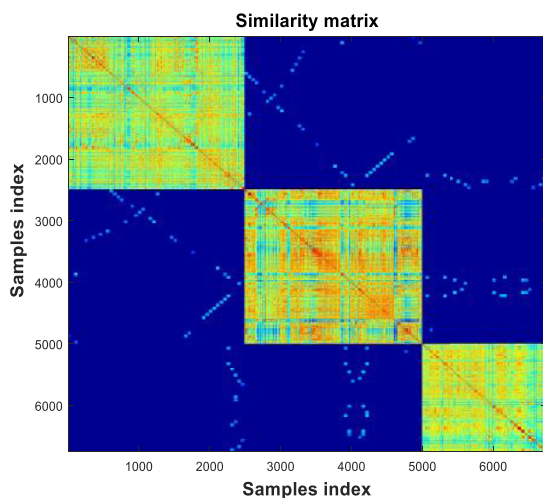
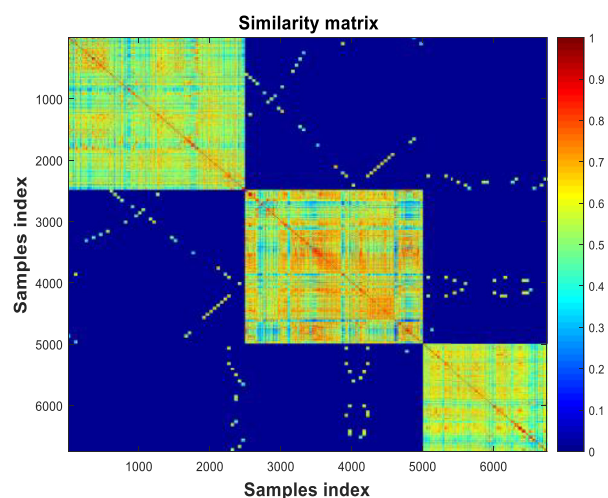
- 1) $\mathbf{S}^{appro \times 0}$ retains the similarity of samples within the same target (i.e., unchanged values), whereas the similarity of samples in different targets is set to zero, as defined in Eq. (6).
- 2) $\mathbf{S}^{appro \times 0.5}$ with each element is presented as

$$S_{i,j}^{appro \times 0.5} = \begin{cases} S_{i,j}^{(c_1, m_1), (c_2, m_2)}, & c_1 = c_2 \\ S_{i,j}^{(c_1, m_1), (c_2, m_2)} \times 0.5, & c_1 \neq c_2 \text{ and } (m_1, m_2) \in m_{appro} \\ 0, & c_1 \neq c_2 \text{ and } (m_1, m_2) \in m_{dis} \end{cases} \quad (33)$$

where $S_{i,j}^{(c_1, m_1), (c_2, m_2)}$ is defined by Eq. (4). Moreover, $\mathbf{S}^{appro \times 0.5}$ retains the similarity of samples within the same target (i.e., unchanged values). Simultaneously, $\mathbf{S}^{appro \times 0.5}$ sets the similarity of samples from approaching aspect-frames in different targets to 0.5 times the original similarity and sets the similarity of those from dissimilar aspect-frames in different targets to zero.

- 3) $\mathbf{S}^{appro \times 1}$ with each element is presented as

$$S_{i,j}^{appro \times 0.5} = \begin{cases} S_{i,j}^{(c_1, m_1), (c_2, m_2)}, & c_1 = c_2 \\ S_{i,j}^{(c_1, m_1), (c_2, m_2)}, & c_1 \neq c_2 \text{ and } (m_1, m_2) \in m_{appro} \\ 0, & c_1 \neq c_2 \text{ and } (m_1, m_2) \in m_{dis} \end{cases} \quad (34)$$

(a) Similarity matrix $S^{appro \times 0}$ (b) Similarity matrix $S^{appro \times 0.5}$ (c) Similarity matrix $S^{appro \times 1}$ **Figs. 5.** Heat map representation of the similarity matrices, where 50 samples correspond to a frame.**Table 2**

Recognition performance with different constraints with 10, 50, and 70 samples per frame.

Number of samples per frame	$S^{appro \times 0}$	$S^{appro \times 0.5}$	$S^{appro \times 1}$
10	74.56	72.44	69.50
50	85.28	84.78	83.89
70	86.17	85.28	84.83

where $S^{appro \times 1}$ retains the similarity of the samples within the same target or from the approaching aspect-frames in different targets (i.e., unchanged values), whereas the similarity of the samples from dissimilar aspect-frames in different targets is set to zero.

Figs. 5(a)–5(c) present the aforementioned three similarity matrices in the form of a heat map. Then, the different SP-MTL models are trained according to the three SP constraints. The recognition results with different constraints are presented in Table 2. The recognition rate of the SP-MTL model with $S^{appro \times 0}$ is higher than those of SP-MTL models with $S^{appro \times 0.5}$ and $S^{appro \times 1}$. Moreover, the recognition rate of SP-MTL model with $S^{appro \times 1}$ is the lowest, thereby indicating that directly constraining the similarity of samples in different targets to zero can improve recognition performance.

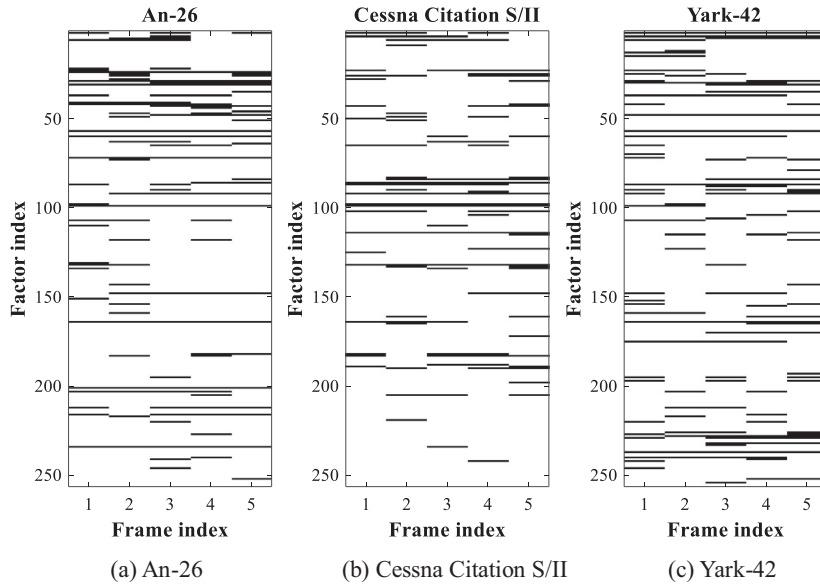


Fig. 6. Posterior mean $\langle u_k^{(c,m)} \rangle$ of the five frames for three targets, with 70 training samples per frame. Black lines, which denote the value of $\langle u_k^{(c,m)} \rangle$, are larger than 0.5.

Table 3
 $ratio_s$ with different training data sizes.

Training data size per aspect-frame	10	30	50	70	90	110
MTL-FA	1.0975	1.0969	1.0957	1.0970	1.0951	1.0960
SP-MTL	1.1048	1.1012	1.0979	1.0990	1.0960	1.0965

The discriminative information can assist in the learning of the generative model by utilizing the SP constraint. A robust constraint on similarity information for different targets can help realize model parameter estimations toward an appropriate direction with limited training data. However, if we consider the information of aspect-angles, that is, slightly retaining the similarity of samples from approaching aspect-frames in different targets, then the expected effect will be weakened. In the case of small-sized training data, the constraint plays an important role in learning. Therefore, a strong constraint, such as $\mathbf{S}^{appro \times 0}$, can achieve better recognition performance than weak constraints, such as $\mathbf{S}^{appro \times 0.5}$ and $\mathbf{S}^{appro \times 1}$. In subsequent experiments, we adopt the setting of similarity matrix $\mathbf{S}^{appro \times 0}$ in all instances.

4.3. Training result analysis

In the SP-MTL model, sub-loading matrices $\mathbf{A}^{(c,m)}$ and $\mathbf{T}^{(c,m)}$ are composed of several columns selected from shared loading matrices $\mathbf{A}$ and $\mathbf{T}$ via binary $z_{i,k}^{(c,m)}$, and $p(z_{i,k}^{(c,m)} = 1) = u_k^{(c,m)}$. The posterior mean $\langle u_k^{(c,m)} \rangle$ of five frames of each target is exhibited in Fig. 6. In this study, K is set to 256, and the training samples are set to 70 per frame. For each target c , if $\langle u_k^{(c,m)} \rangle$ is larger than 0.5, then we consider the latent factor k suitable for frame m , as illustrated by the black lines in Fig. 6. Only approximately 50% may be selected for each frame, thereby confirming the sparsity of binary vector $z_{i,k}^{(c,m)}$. Moreover, the columns of $\mathbf{A}$ that are used to construct observed data for the same target are similar, thus indicating that the statistical models for different frames share many common latent factors to represent the original signal for each target. This condition is beneficial for recognition performance.

To verify the performance of similarity preservation, $ratio_s$ with different training data sizes (Table 3) is defined as follows:

$$ratio_s = \frac{\tilde{s}_{intra}}{\tilde{s}_{inter}}, \quad (35)$$

where $\tilde{s}_{intra}$ denotes the average similarity of training data reconstructions within the same target, and $\tilde{s}_{inter}$ denotes the average similarity of training data reconstructions in different targets. In Table 4, $ratio_d$ with different training data sizes is defined as follows:

$$ratio_d = \frac{\tilde{d}_{inter}}{\tilde{d}_{intra}}, \quad (36)$$

Table 4
 $ratio_d$ with different training data sizes.

Training data size per aspect-frame	10	30	50	70	90	110
MTL-FA	1.3703	1.4790	1.4981	1.6264	1.6526	1.6943
SP-MTL	1.4174	1.6279	1.6544	1.6937	1.7239	1.7284

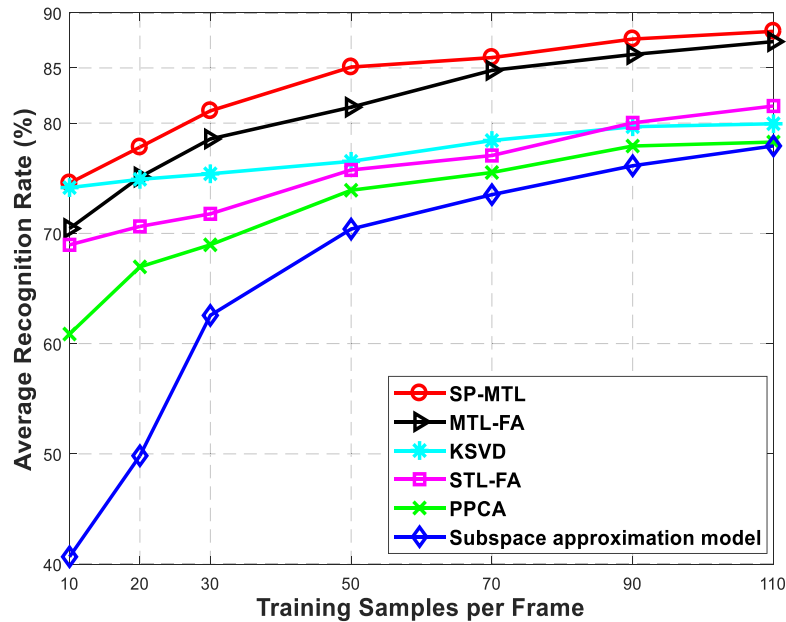


Fig. 7. Variations in recognition accuracies with training samples per frame of the SP-MTL, MTL-FA, K-SVD, STL-FA, PPCA, and subspace approximation models.

where $\tilde{d}_{\text{inter}}$ denotes the average KL distance of training data reconstructions in different targets, and $\tilde{d}_{\text{intra}}$ denotes the average KL distance of training data reconstructions within the same target. As discussed in Section 2.2, large values of $ratio_s$ and $ratio_d$ strengthen the discriminative capability of the learned statistical model. In Tables 3 and 4, in which training samples per frame vary from 10 to 110, the values of $ratio_s$ and $ratio_d$ are larger in the proposed SP-MTL model than in the MTL-FA model. This result validates the improvement of the discriminative capability of the proposed model after adding the SP constraint.

4.4. Comparison with related models

The recognition accuracies of our proposed SP-MTL, MTL-FA [11], STL-FA [8], K-SVD, PPCA [8,27] and subspace approximation models [8] against the size of the training data are shown in Fig. 7. Here, K-SVD uses the K-SVD algorithm [1], which is a dictionary learning method based on optimization, for feature extraction and uses linear support vector machine as the classifier for extracted features. In Fig. 7, the recognition rate of each model is improved with an increase in the number of training samples per frame. The recognition performance of the SP-MTL model is constantly superior to those of other models while the training samples per frame vary from 10 to 110. Furthermore, the recognition rate of SP-MTL model is significantly higher than those of the other models when the training samples per frame are less than 70. The SP-MTL model is also slightly better than the MTL-FA model when the training data size per frame is between 70 and 110. If we assume that the average recognition rate threshold is 75%, then the SP-MTL model can work well with 11 training samples for each frame. By contrast, the MTL-FA, K-SVD, STL-FA, PPCA, and subspace approximation models require approximately 20, 30, 50, 70, and 90 training samples per frame, respectively. If we increase the recognition rate to over 80% and use this value as a benchmark to determine the training samples per frame, then the SP-MTL model only requires 26 training samples per frame. The MTL-FA and STL-FA models require approximately 40 and 90 samples per frame, correspondingly. Therefore, the training data size that is required by the MTL-FA and STL-FA models to obtain the same recognition performance is nearly twice or thrice that of the SP-MTL model. The findings verified that the SP-MTL is desirable for solving a problem with small-sized training data. However, the improvement of the recognition performance of the SP-MTL model slows down when the number of training samples per frame increases, as expected. The reason for this phenomenon is as follows. If sufficient training data are available, then the learned generative model can precisely represent the distribution of

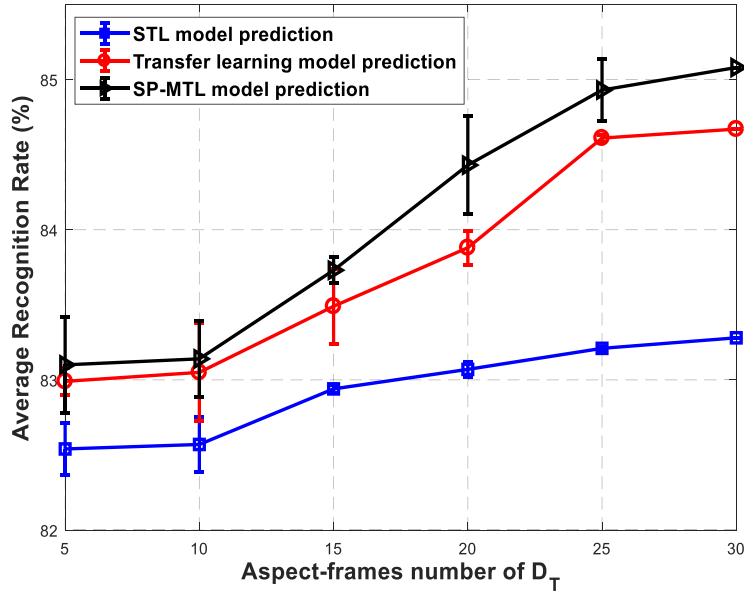


Fig. 8. Recognition accuracies of the different model prediction methods vs. the number of aspect-frames in D_T . The number of training samples per aspect-frame is 50, and recognition rates are averaged over five runs. Error bars denote the corresponding standard deviations.

Table 5

Computation costs of model prediction methods with one aspect-frame of D_T .

Model prediction methods	STL model prediction	Transfer learning model prediction	SP-MTL model prediction
Computational time cost (min)	0.33	0.83	326

data. However, the MTL-FA model is incapable of precisely representing the data when the amount of training data is small. We drive the model parameters of the MTL-FA model to move forward to an optimal direction by adding the SP constraint.

4.5. Performance analysis of model prediction

To verify the recognition performance of model prediction based on transfer learning, we assume that the original training data D_S include the following: 1st–40th frames in An-26 data, 1st–40th frames in Cessna Citation S/II data, and 1st–25th frames in Yark-42 data. The remaining frames are assumed to be new data in D_T for model prediction (i.e., 41st–50th frames in the An-26 data, 41st–50th frames in the Cessna Citation S/II data, and 26th–35th frames in the Yark-42 data). The recognition accuracies of the different model prediction methods in relation to the number of aspect-frames in D_T are shown in Fig. 8. Two model prediction methods are adopted for comparison. First, all the following methods adopt the SP-MTL model for model training in T_S . Then, “STL model prediction” uses the STL-FA model in constructing the statistical model for each aspect-frame in D_T , “SP-MTL model prediction” combines D_S and D_T to rebuild the statistical models for all data, and “transfer learning model prediction” denotes the proposed model prediction method.

In Fig. 8, the recognition performance of the “STL model prediction” obtains the lowest accuracy. The proposed “transfer learning model prediction” exhibits a performance similar to that of the “SP-MTL model prediction,” thereby demonstrating that the transferred knowledge of T_S plays an important role in model prediction for a small amount of new data. Table 5 presents the computational time cost of the different model prediction methods with only one aspect-frame in D_T . Compared with the “SP-MTL model prediction,” the computational time cost of the “transfer learning model prediction” is reduced by approximately 99.75%, as listed in Table 5. In summary, the proposed “transfer learning model prediction” significantly reduces the computation cost compared with “SP-MTL model prediction,” and is more accurate than the “STL model prediction” in terms of recognition performance. Therefore, the “transfer learning model prediction” can achieve an appropriate balance between computational time cost and recognition performance in real applications.

5. Conclusion

We propose a HRRP target recognition method based on the SP-MTL model. MTL and SP constraint are adopted to solve the problem of estimating frame-condition PDFs with small-sized training data. Then, the observed data are modeled with BPFA. Transfer learning theory is used for model prediction with a small amount of data obtained in real time. Experiments on the measured data show that the method that incorporates the MTL and SP proposed in this study can exhibit improved

recognition performance when the size of the training data is small. Furthermore, applying transfer learning theory improves the speed of learning statistical models of new data obtained in real time.

Acknowledgments

This work was partially supported by the National Science Foundation of China (No. 61771362 and No. 61671354).

References

- [1] M. Aharon, M. Elad, A. Bruckstein, K-SVD: an algorithm for designing overcomplete dictionaries for sparse representation, *IEEE Trans. Signal Process.* 54 (2006) 4311–4322.
- [2] Z. Alhalah, L. Rybok, R. Stiefelbogen, What to transfer? High-level semantics in transfer metric learning for action similarity, in: *Proceedings of the International Conference on Pattern Recognition*, 2014, pp. 2775–2780.
- [3] Y. Anzai, *Pattern Recognition and Machine Learning*, Academic Press, 2006.
- [4] M.J. Beal, *Variational Algorithms For Approximate Bayesian Inference*, University College London, 2003.
- [5] B. Chen, H. Liu, J. Chai, Z. Bao, Large margin feature weighting method via linear programming, *IEEE Trans. Knowl. Data Eng.* 21 (2009) 1475–1488.
- [6] K. Copsey, A. Webb, Bayesian gamma mixture model approach to radar target recognition, *IEEE Trans. Aerosp. Electron. Syst.* 39 (2004) 1201–1217.
- [7] L. Du, M. Chen, J. Lucas, L. Carin, Sticky hidden Markov modeling of comparative genomic hybridization, *IEEE Trans. Signal Process.* 58 (2010) 5353–5368.
- [8] L. Du, H. Liu, Z. Bao, Radar HRRP statistical recognition: parametric model and model selection, *IEEE Trans. Signal Process.* 56 (2008) 1931–1944.
- [9] L. Du, H. Liu, Z. Bao, M. Xing, Radar HRRP target recognition based on higher order spectra, *IEEE Trans. Signal Process.* 53 (2005) 2359–2368.
- [10] L. Du, H. Liu, Z. Bao, J. Zhang, A two-distribution compounded statistical model for radar HRRP target recognition, *IEEE Trans. Signal Process.* 54 (2006) 2226–2238.
- [11] L. Du, H. Liu, P. Wang, B. Feng, M. Pan, Z. Bao, Noise robust radar HRRP target recognition based on multitask factor analysis with small training data size, *IEEE Trans. Signal Process.* 60 (2012) 3546–3559.
- [12] L. Du, P. Wang, L. Zhang, H. He, H. Liu, Robust statistical recognition and reconstruction scheme based on hierarchical Bayesian learning of HRR radar target signal, *Expert Syst. Appl.* 42 (2015) 5860–5873.
- [13] W.R.E. Gilks, S.R. Richardson, D.J.E. Spiegelhalter, Markov chain Monte Carlo in practice, *Technometrics* 92 (1996).
- [14] L. Hong, N. Cui, M.T. Pronobis, S. Scott, Simultaneous ground moving target tracking and identification using wavelets features from HRR data ☆, *Inf. Sci.* 162 (2004) 249–274.
- [15] S.P. Jacobs, *Automatic Target Recognition Using High-Resolution Radar Range-Profiles*, Washington University, 1997.
- [16] Z. Jiang, Z. Lin, L.S. Davis, Label consistent K-SVD: learning a discriminative dictionary for recognition, *IEEE Trans. Pattern Anal. Mach. Intell.* 35 (2013) 2651–2664.
- [17] K.T. Kim, D.K. Seo, H.T. Kim, Efficient classification of ISAR images, *IEEE Trans. Antennas Prop.* 53 (2005) 1611–1621.
- [18] A. Naik, A. Charuvaka, H. Rangwala, Classifying Documents within multiple hierarchical datasets using multi-task learning, in: *Proceedings of the IEEE International Conference on TOOLS with Artificial Intelligence*, 2013, pp. 390–397.
- [19] D. Osaku, R.Y.M. Nakamura, L.A.M. Pereira, R.J. Pisani, A.L.M. Leveda, F.A.M. Cappabianco, A.X. Falcão, J.P. Papa, Improving land cover classification through contextual-based optimum-path forest, *Inf. Sci.* 324 (2015) 60–87.
- [20] J. Paisley, L. Carin, Nonparametric factor analysis with beta process priors, in: *Proceedings of the International Conference on Machine Learning*, 2009, pp. 777–784.
- [21] M. Pan, Multi-task hidden Markov modeling of spectrogram feature from radar high-resolution range profiles, *Eur. J. Adv. Signal Process.* (2012) 86 2012.
- [22] S.J. Pan, Q. Yang, *A Survey on Transfer Learning*, IEEE Educational Activities Department, 2010.
- [23] G. Parisi, J. Machta, Statistical field theory, *Phy. Today* 41 (1988) 110.
- [24] M.T. Rosenstein, Z. Marx, L.P. Kaelbling, T.G. Dietterich, To Transfer or not to transfer, in: 2005, pp. S20.
- [25] D.B. Rubin, D.T. Thayer, EM algorithms for ML factor analysis, *Psychometrika* 47 (1982) 69–76.
- [26] D. Silver, I. Guyon, G. Taylor, V. Lemaire, G. Dror, ICML2011 unsupervised and transfer learning workshop, *J. Mach. Learn. Res.* 27 (2012) 1–16.
- [27] M.E. Tipping, C.M. Bishop, Mixtures of principal component analyzers, *Neural Comput.* 11 (1997) 443–482.
- [28] A. Toumi, A. Khenchaf, B. Hoeltzener, A retrieval system from inverse synthetic aperture radar images: application to radar target recognition, *Inf. Sci.* 196 (2012) 73–96.
- [29] A.R. Webb, *Statistical pattern recognition*, 2003.
- [30] D. Xu, L. Du, H. Liu, P. Wang, J. Yan, Y. Cong, X. Han, Compressive sensing of stepped-frequency radar based on transfer learning, *IEEE Trans. Signal Process.* 63 (2015) 3076–3087.
- [31] Y. Xue, X. Liao, L. Carin, B. Krishnapuram, Multi-task learning for classification with Dirichlet process priors, *J. Mach. Learn. Res.* 8 (2007) 35–63.
- [32] Q. Zhao, J.C. Principe, Support vector machines for SAR automatic target recognition, *IEEE Trans. Aerosp. Electron. Syst.* 37 (2001) 643–654.
- [33] J. Zhou, Z. Shi, C. Xiao, F. Qiang, Automatic target recognition of SAR images based on global scattering center model, *IEEE Trans. Geosci. Remote Sens.* 49 (2011) 3713–3729.